ARMAN BABAEI

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PROFESSIONAL SUMMARY

Platform and distributed-systems engineer with production experience across backend development, SRE, and platform engineering. Builds reliability and deployment tooling for Kubernetes-based microservices and brings research experience in distributed protocols, programmable networking, and blockchain systems.

SKILLS

Languages	Go, Python, C++, Java, Kotlin, JavaScript, Rust, Bash
Platform/Cloud	Kubernetes, Docker, Helm, Flagger, Istio, ArgoCD/Flux, GitLab CI
Observability	Prometheus/VictoriaMetrics, Grafana, Graylog, Sentry
Backend/Data	gRPC, Kafka, Django, Spring, Node.js, PostgreSQL, MySQL, MongoDB, Redis
Systems	Distributed systems, reliability engineering, canary delivery, P4, Linux networking

WORK EXPERIENCE

Tapsi September 2024 – Present
Proficient Platform Engineer *Tehran, Iran*

- **Platform Engineer (Apr'26 – Present):** Build and roll out safer Kubernetes delivery workflows with Helm, Flagger, Istio, and GitOps; develop platform automation for canary releases, rollback safety, observability, and developer experience across production services.
- **SRE (Sep'25 – Apr'26):** Improved service reliability through production monitoring, incident analysis, on-call operations, and debugging across microservices, service-mesh traffic, and deployment infrastructure.
- **Backend Engineer (Sep'24 – Sep'25):** Built production passenger–driver matching features; extracted and rewrote services from Node.js to Go, including services handling roughly 1k requests/second; maintained Go/Python/Node.js services using Kafka and Kubernetes.

Undisclosed Hedge Fund September 2023 – February 2024
Full-stack Developer Intern *Remote*

- Developed a Python/Django job-scheduler web service for intra-company tasks, including backend REST APIs and frontend integration.

Mahsan Co. September 2020 – July 2022
C++/Python Developer *Tehran, Iran*

- Led development of a Python domain-specific language for end-to-end test automation; developed parallel network-relaying software in C++; served as Scrum Master for a team of five.

Allameh Helli High School June 2019 – October 2020
Algorithm Design Teacher *Tehran, Iran*

- Taught algorithm design to students preparing for Iran's National Olympiad in Informatics.

EDUCATION

Sharif University of Technology September 2024 – Expected August 2026
M.S. in Computer Engineering (Networks)

- Current GPA: 18.50/20; TOEFL: 113/120

Sharif University of Technology September 2019 – July 2024
B.S. in Computer Engineering

· Overall GPA: 19.48/20; Rank: 15/190

HONORS

2026 Ranked **#7** in Iran's National Doctoral Entrance Exam (Networks and Systems subgroup)
2024 Ranked **#4** in Iran's Master University Entrance Exam (Systems and Networks subgroup)
2019 Ranked top **0.01%** in Iran's National University Entrance Exam
2018 Bronze medalist, Iran National Olympiad in Informatics (INOI)

SELECTED RESEARCH EXPERIENCE

Sharif University of Technology August 2025 – Present
L4S/AQM Prototyping for Volumetric Video Streaming *Tehran, Iran*

· Prototype DualPI2/L4S low-latency AQM in P4_16 (BMv2), with per-egress-port state and traffic classification; compare against Linux implementations and evaluate queueing behavior for latency-sensitive media.

Stony Brook University October 2024 – Present
Adaptive Leader Election for PBFT *Remote*

· Design a modular PBFT-style leader-election component and adaptive feedback loop using Byzantine-robust aggregation of node progress signals.

EPFL DEDIS July 2023 – September 2023
Fairness in Decentralized Storage Systems *Lausanne, Switzerland*

· Built a mathematical model for fairness in Swarm, extended a Go simulator with churn and heterogeneous node behavior, and designed a network-shuffling mechanism that improved fairness. (arXiv)

ISTA SPiDerS July 2022 – February 2023
Parallel Execution in Blockchain Protocols *Vienna, Austria*

· Reproduced prior results, built the experimental test bench, analyzed commercial-blockchain workloads, and contributed implementation work for parallel smart-contract execution. (arXiv)

PUBLICATIONS

Pythia: Supercharging Parallel Smart Contract Execution to Guide Stragglers and Full Nodes to Safety SIB – 2024
Altruism, reciprocity, and tokens to reward forwarding data: Is that fair? ICBC – 2024